

# Lecture 8: Motion

# Today:

**00:00)** Optical flow,

**00:15)** Lucas-Kanade, aperture, corners

**00:45)** ---- Break ----

**01:00)** Optical flow pipeline

**01:45)** ---- Break ----

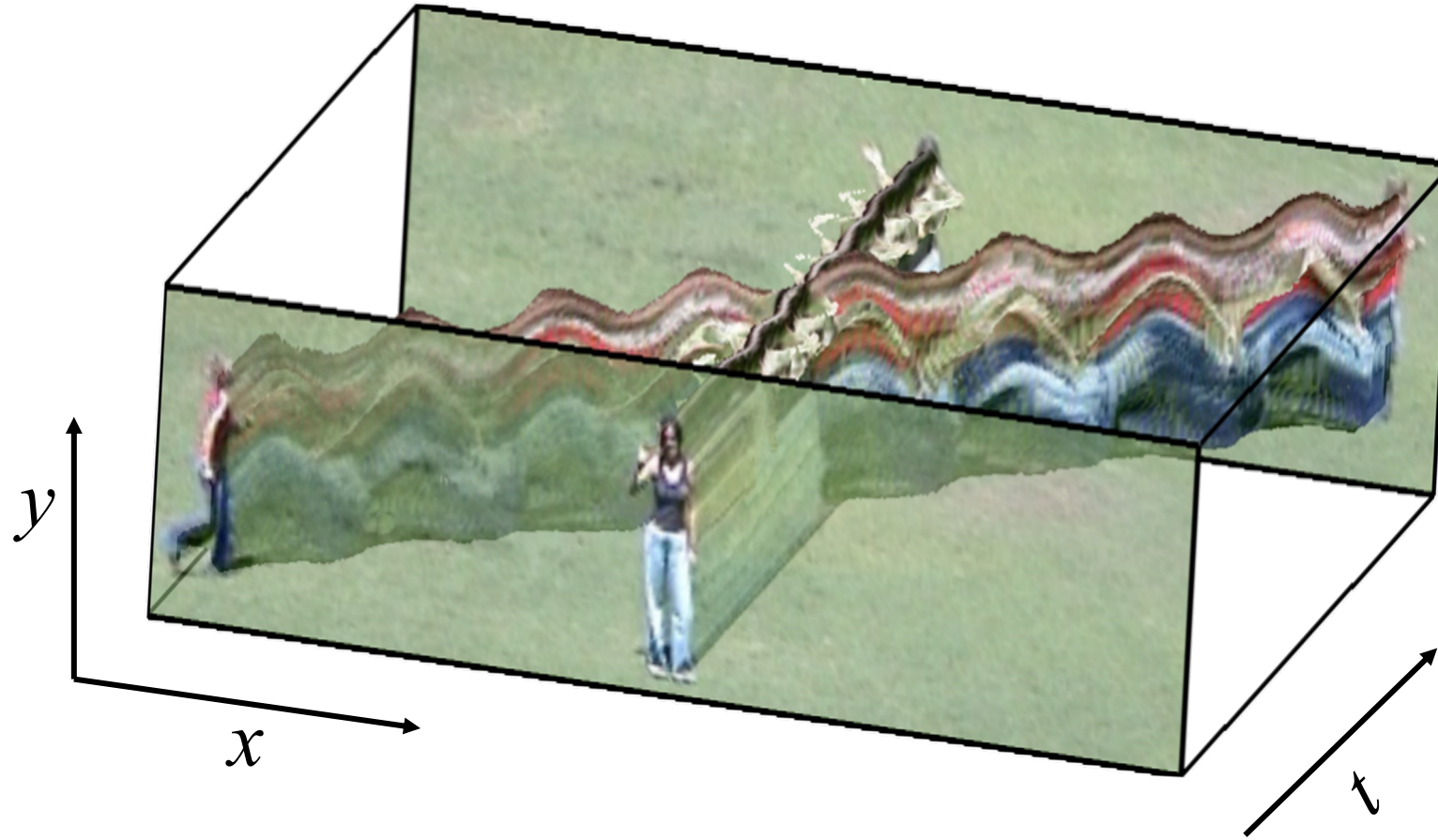
**02:00)** Global motion & segmentation

**02:20)** Deep optical flow

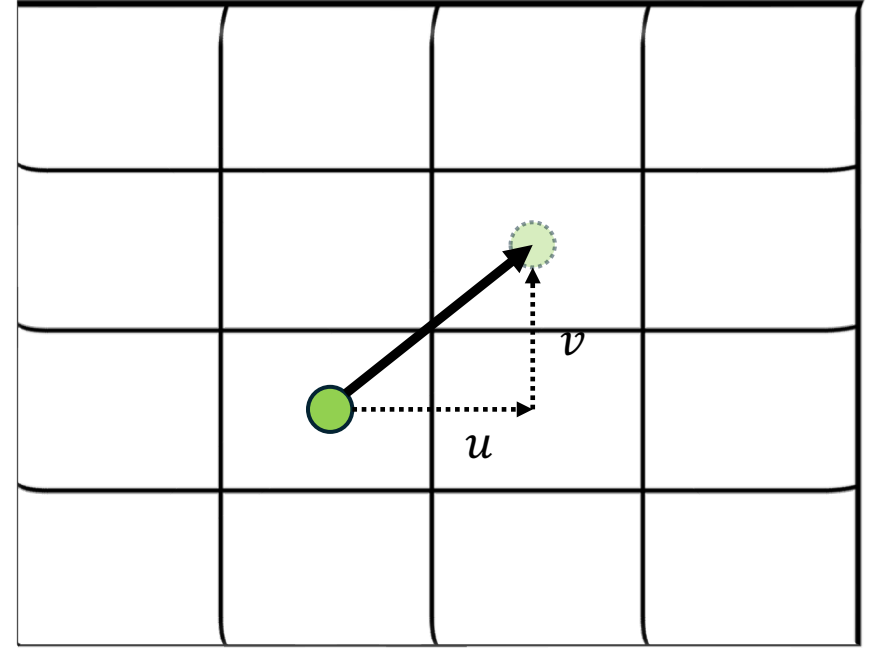
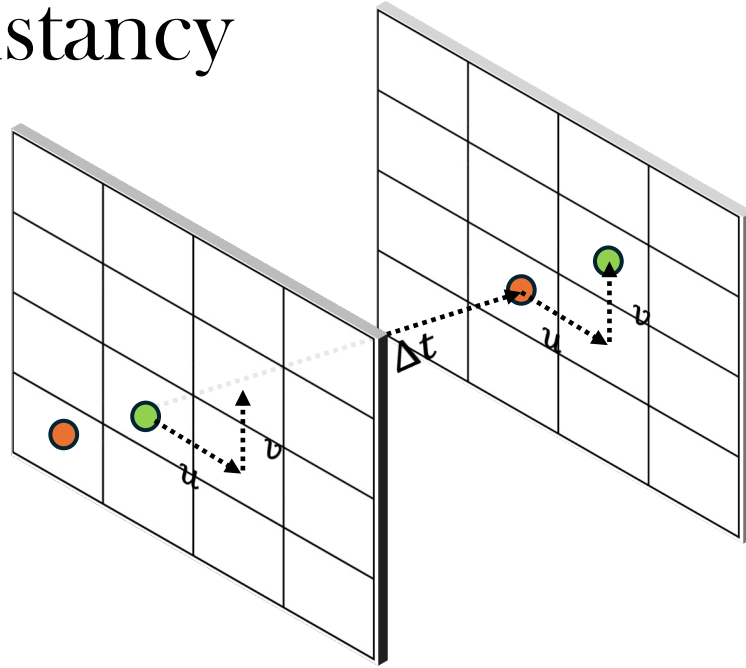
**02:45)** ---- Break ----

**03:00)** Tutorial: Detection and Segmentation

# A video is a space-time volume



# Brightness constancy



$$I(x, y, t) = I(x + u, y + v, t + \Delta t) \approx I(x, y, t) + \frac{dI}{dx}u + \frac{dI}{dy}v + \frac{dI}{dt}\Delta t$$

$$I_x u + I_y v + I_t = 0$$

$$(I_x, I_y) \begin{pmatrix} u \\ v \end{pmatrix} = -I_t$$

$$\nabla I^T V = -I_t$$

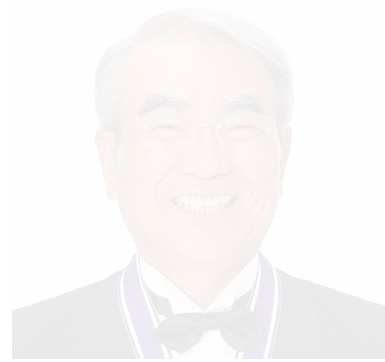
2 problems/questions:

1. One equation, two variables! 🧐
2. What do **spatial** have to do with **temporal** motion?

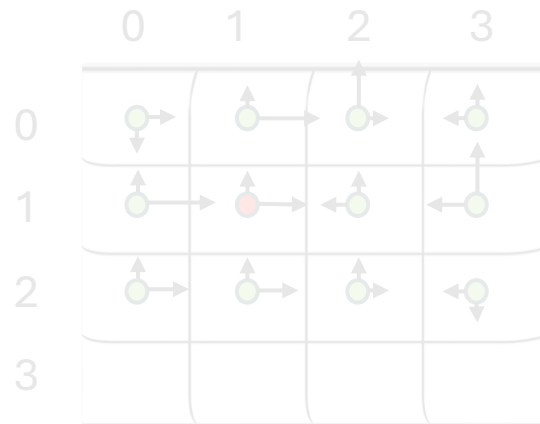
# Lucas-Kanade

Assume close pixels have the same flow.

**Only from the perspective of a local pixel each time!**



Takeo Kanade



$$\underbrace{\begin{pmatrix} I_x^{(0,0)} & I_y^{(0,0)} \\ I_x^{(0,1)} & I_y^{(0,1)} \\ \vdots & \vdots \\ I_x^{(2,2)} & I_y^{(2,2)} \end{pmatrix}}_A \underbrace{\begin{pmatrix} u \\ v \end{pmatrix}}_x = \underbrace{\begin{pmatrix} -I_t^{(0,0)} \\ -I_t^{(0,1)} \\ \vdots \\ -I_t^{(2,2)} \end{pmatrix}}_y$$

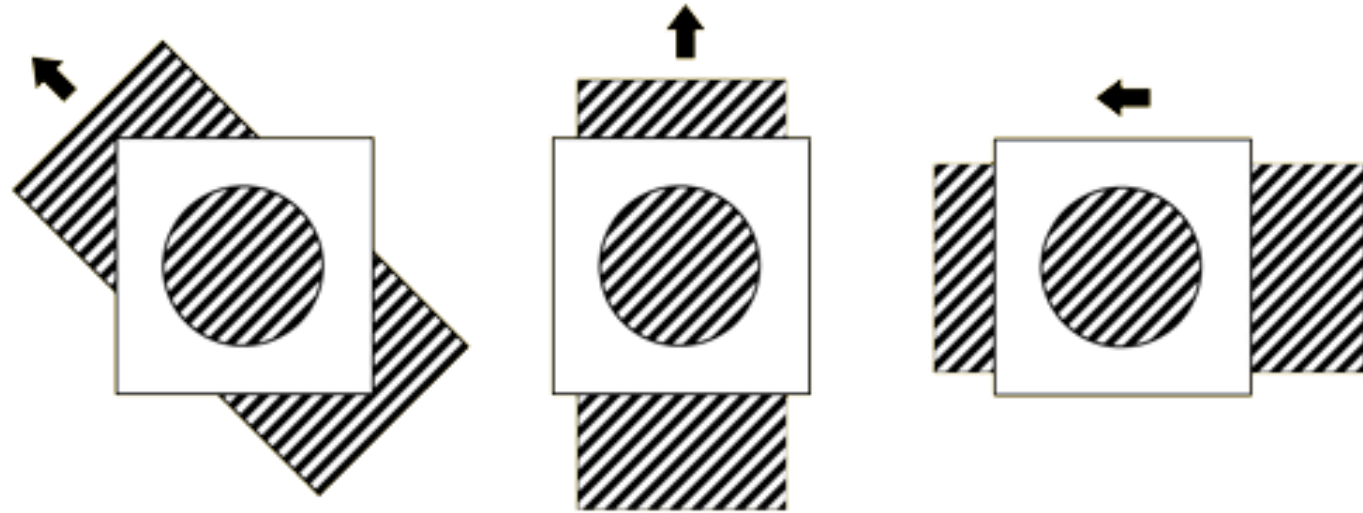
$$x = (A^T A)^{-1} A^T y \quad \longrightarrow$$

$$\underbrace{\begin{pmatrix} u \\ v \end{pmatrix}}_x = \underbrace{\begin{pmatrix} \sum_i I_x^{(i)2} & \sum_i I_x^{(i)} I_y^{(i)} \\ \sum_i I_x^{(i)} I_y^{(i)} & \sum_i I_y^{(i)2} \end{pmatrix}^{-1}}_{(A^T A)^{-1}} \underbrace{\begin{pmatrix} -\sum_i I_x^{(i)} I_t^{(i)} \\ -\sum_i I_y^{(i)} I_t^{(i)} \end{pmatrix}}_{A^T y}$$

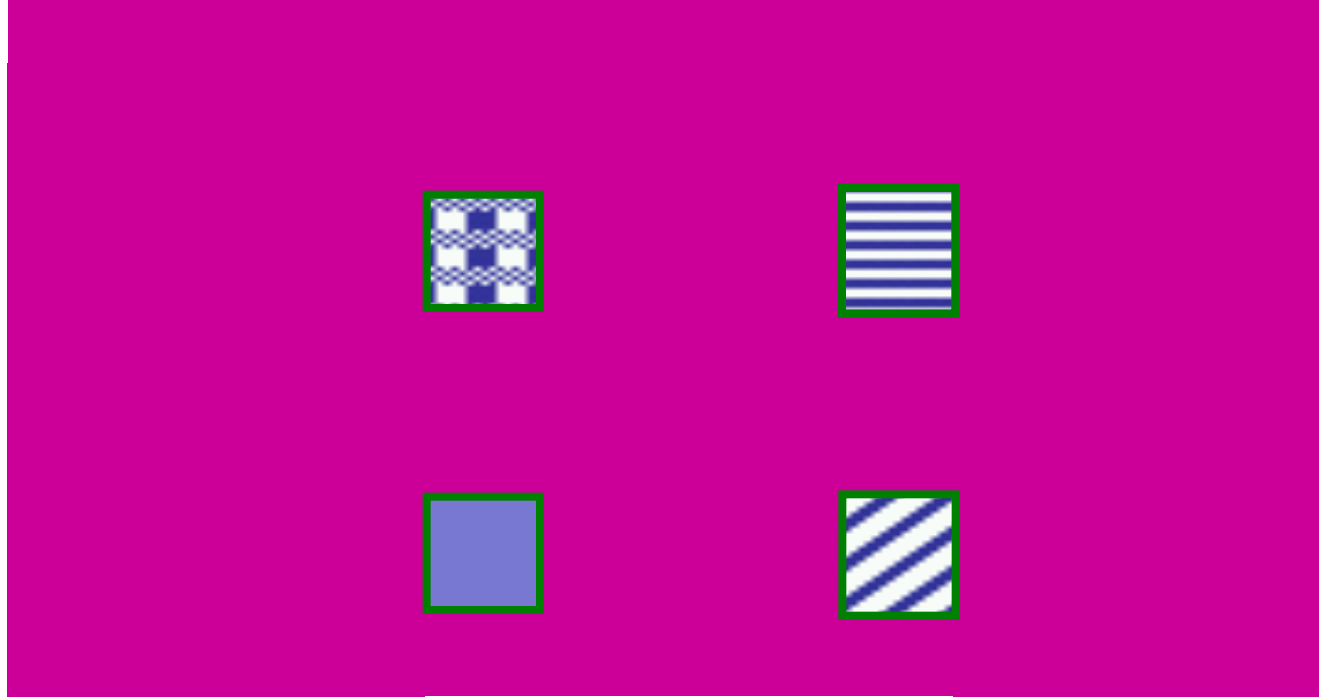
## Questions:

1. What if  $A^T A$  is singular?
2. What physically would that mean?
3. What do barbers ✂ have to do with all of that?

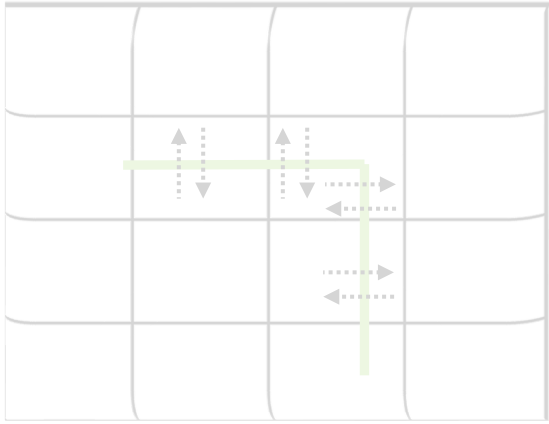
# The aperture problem



# The aperture problem



# Corner detection:



(Assume only vertical or horizontal gradients exist.)

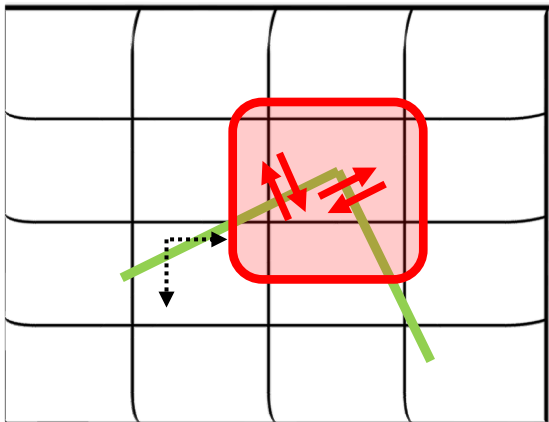
variance of derivatives:

$$\Sigma \left( \frac{dI}{dx} \right)^2$$

$$\Sigma \left( \frac{dI}{dy} \right)^2$$

$$A = \begin{pmatrix} I_x^{(0,0)}, I_y^{(0,0)} \\ I_x^{(0,1)}, I_y^{(0,1)} \\ \vdots \\ I_x^{(2,2)}, I_y^{(2,2)} \end{pmatrix}$$

What about diagonals?



PCA = rotate the axes to strongest variances directions!

$$PCA \vec{\nabla} I \text{ over patch} \rightarrow A^T A = \begin{pmatrix} \Sigma I_x^2 & \Sigma I_x I_y \\ \Sigma I_x I_y & \Sigma I_y^2 \end{pmatrix} = U \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} V^T$$

$$\begin{cases} \text{Smooth} & \lambda_1, \lambda_2 \ll 1 \\ \text{Edge} & \lambda_1 \gg \lambda_2 \\ \text{Corner} & \lambda_1, \lambda_2 \gg 0 \end{cases}$$

$I_x^2$  and  $I_y^2$  in the rotated coordinates

# Back to optical flow:

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \sum_i I_x^{(i)2} & \sum_i I_x^{(i)} I_y^{(i)} \\ \sum_i I_x^{(i)} I_y^{(i)} & \sum_i I_y^{(i)2} \end{pmatrix}^{-1} \begin{pmatrix} -\sum_i I_x^{(i)} I_t^{(i)} \\ -\sum_i I_y^{(i)} I_t^{(i)} \end{pmatrix} = \boxed{V \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}^{-1} U^T \begin{pmatrix} -\sum_i I_x^{(i)} I_t^{(i)} \\ -\sum_i I_y^{(i)} I_t^{(i)} \end{pmatrix}}$$

## **Harris (1988):**

$$R = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2) \quad (k \approx 0.04 - 0.06)$$

$$R = \det(M) - k \cdot \text{tr}(M)$$

Advantage: Fast (avoids eigen decomp)

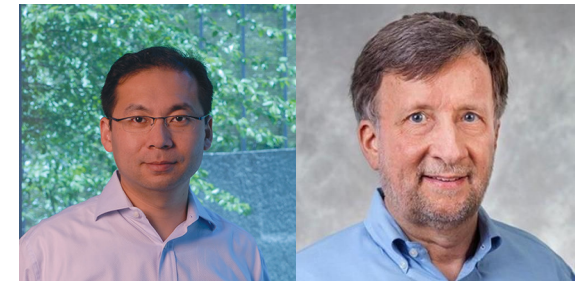


Chris Harris

## **Shi-Tomasi (1993):**

$$R = \lambda_2 \quad \{= \min(\lambda_1, \lambda_2)\}$$

Advantage: scores high only if both eigen vals are high.

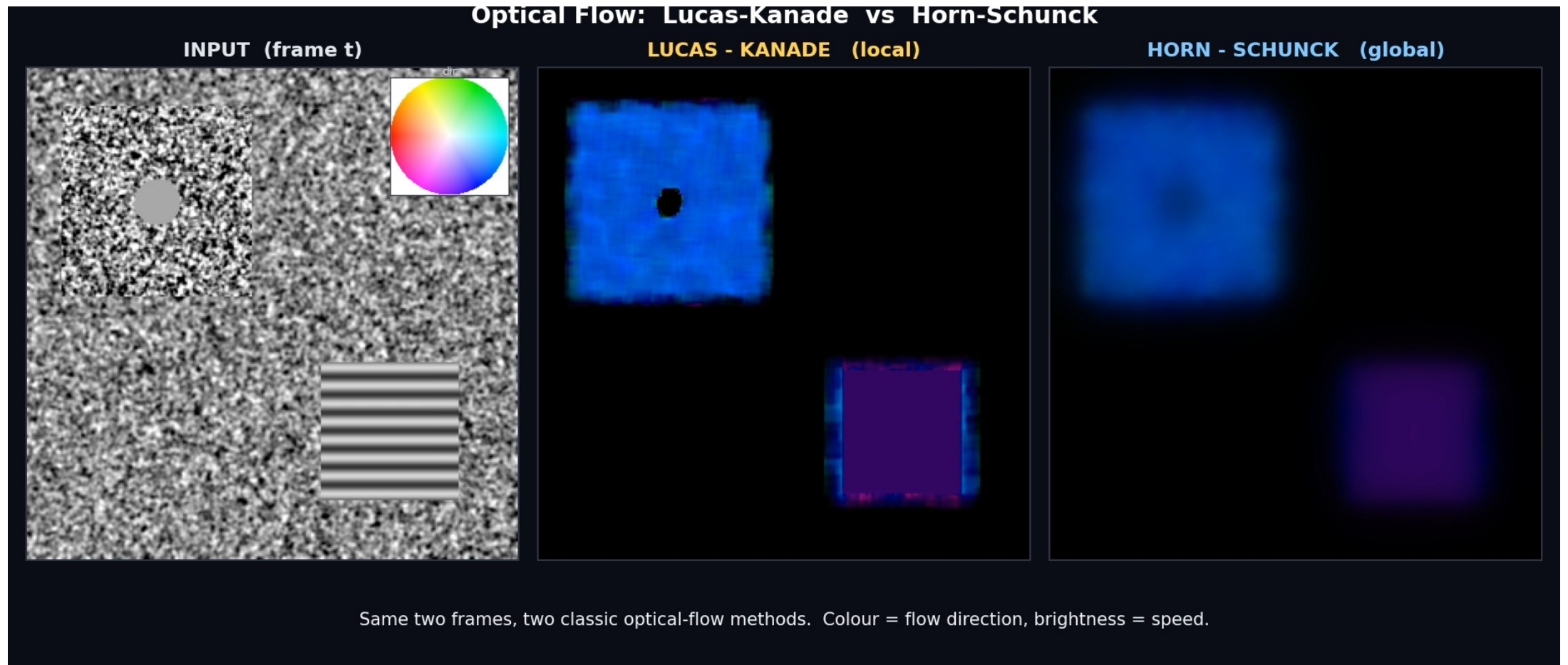


Jianbo Shi Carlo Tomasi

# Horn-Schunck

$$E = \int \left( (I_x u + I_y v + I_t)^2 + \alpha^2 (\|\vec{\nabla} u\|^2 + \|\vec{\nabla} v\|^2) \right) dx dy$$

- Enforce smoothness globally.
- Dense flow everywhere, even when no spatial grads.
- Smoothing over edges



# 2 problems

1. Only first order approx.

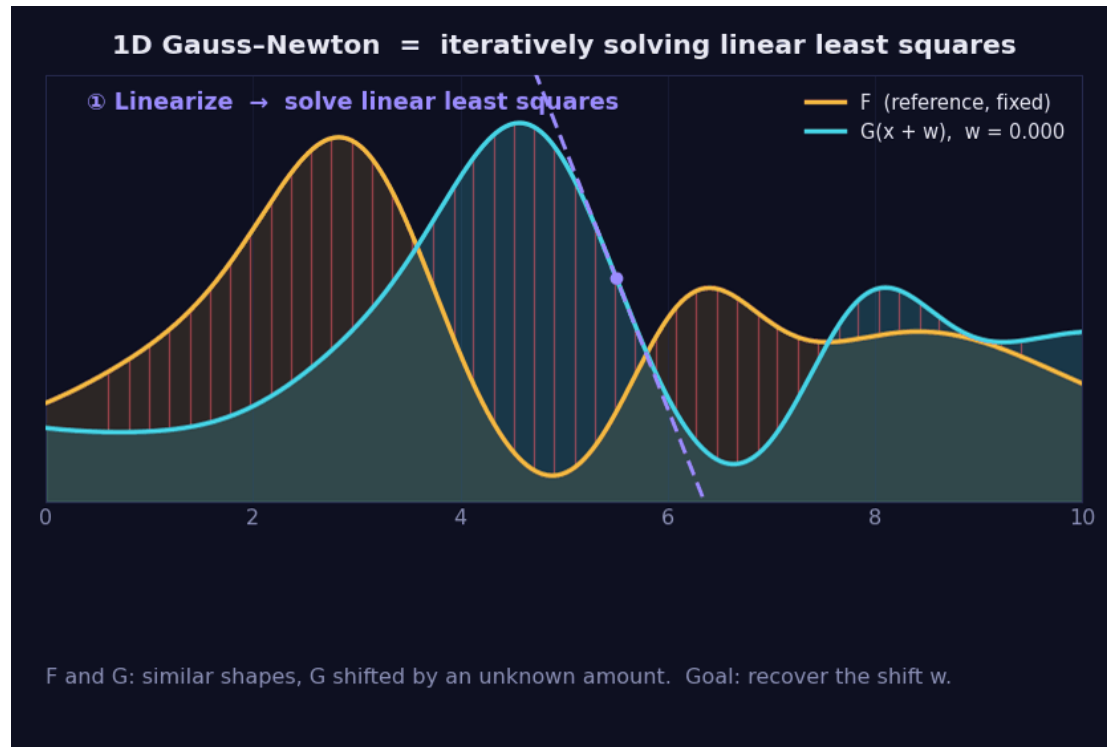
$$I(x + u, y + v, t + \Delta t) \approx I(x, y, t) + I_x u + I_y v + I_t \Delta t$$

2. Only good for < 1px shift.

$$I_x \Big|_{x_0, y_0} = I_{x_0+1, y_0} - I_{x_0, y_0}$$

# Beyond first order solution: Gauss-Newton

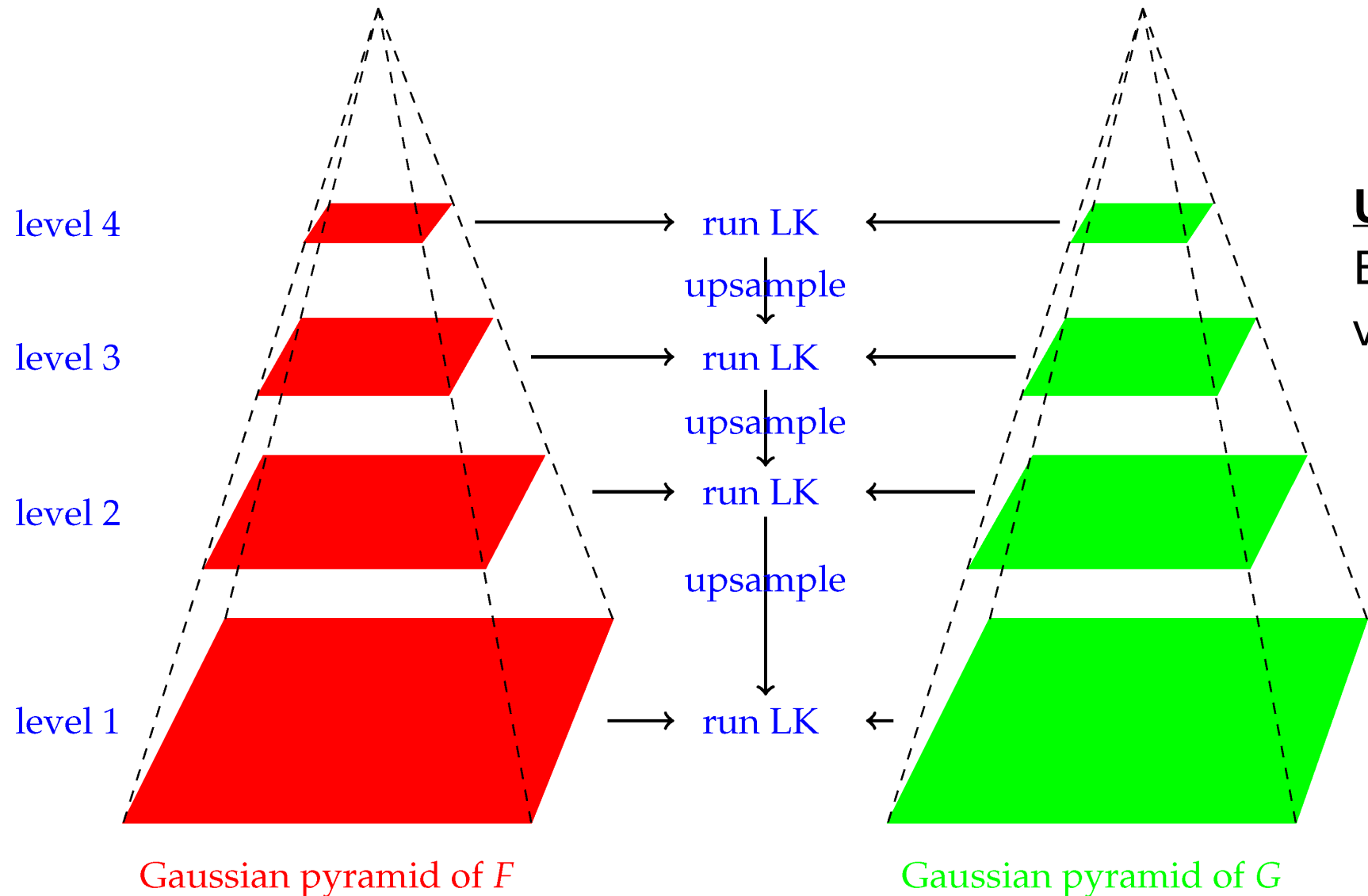
Solve non-linear Least Squares, using linear Least Squares iteratively



Repeat:

- Calc flow field
- Backward warp frame  $t+1 \rightarrow t$
- (now you have residual)

# What about bigger motion?

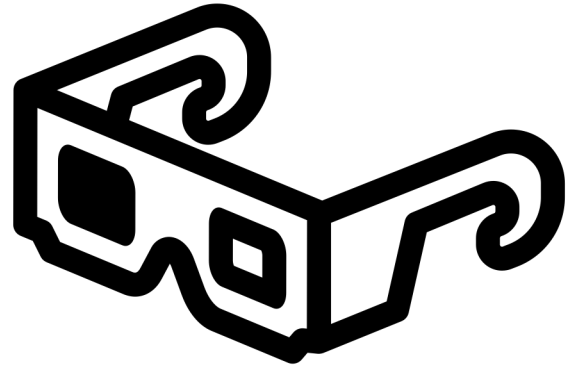


## Upsample note:

Both the locations and the vector sizes!

# Optical flow pipeline

Demo:



# Global motion modeling

## Before:

Each pixel by itself (with global regularization).

$u, v$  are the parameters to optimize.

## Now:

One rule for all pixels, conditioned on their location.

$$u(x, y; a_0 \dots a_{N/2})$$

$$v(x, y; a_{N/2} \dots a_N)$$

E.g.,

$$u(x, y) = a_0 + a_1x + a_2y$$

$$v(x, y) = a_3 + a_4x + a_5y$$

(Affine)

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} a_1 & a_2 \\ a_4 & a_5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a_0 \\ a_3 \end{pmatrix}$$

# Global motion modeling

Before:

$$I_x u + I_y v + I_t = 0$$

Now (E.g., affine):

$$u(x, y) = a_0 + a_1 x + a_2 y$$

$$v(x, y) = a_3 + a_4 x + a_5 y$$

$$I_x(a_0 + a_1 x + a_2 y) + I_y(a_3 + a_4 x + a_5 y) + I_t = 0$$

6 unknowns. Over all pixels in the image.

$$\begin{pmatrix} I_{x_1} & x_1 I_{x_1} & y_1 I_{x_1} & I_{y_1} & x_1 I_{y_1} & y_1 I_{y_1} \\ I_{x_2} & x_2 I_{x_2} & y_2 I_{x_2} & I_{y_2} & x_2 I_{y_2} & y_2 I_{y_2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{pmatrix} = - \begin{pmatrix} I_t(x_1, y_1) \\ I_t(x_2, y_2) \\ \vdots \end{pmatrix}$$

Solve with Least-Squares

# Homography $\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} \propto H \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} x' - x \\ y' - y \end{pmatrix} = \frac{H \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}}{H \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}_z} - \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{h_{31}x + h_{32}y + 1} \begin{pmatrix} h_{11}x + h_{12}y + h_{13} \\ h_{21}x + h_{22}y + 1 \end{pmatrix} - \begin{pmatrix} x \\ y \end{pmatrix}$$

$$I_x \left( \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + 1} - x \right) + I_y \left( \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + 1} - y \right) + I_t = 0$$

$$\begin{pmatrix} x_1 I_{y_1} & y_1 I_{x_1} & I_{x_1} & x_1 I_{y_1} & y_1 I_{y_1} & I_{y_1} & I_t^1 x_1 - I_x^1 (x_1)^2 - I_y^1 x_1 y_1 & I_t^1 y_1 - I_x^1 x_1 y_1 - I_y^1 (y_1)^2 \\ & & & & & & \vdots & \end{pmatrix} \begin{pmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{pmatrix} = - \begin{pmatrix} I_t^1 - I_x^1 x_1 - I_y^1 y_1 \\ \vdots \end{pmatrix}$$

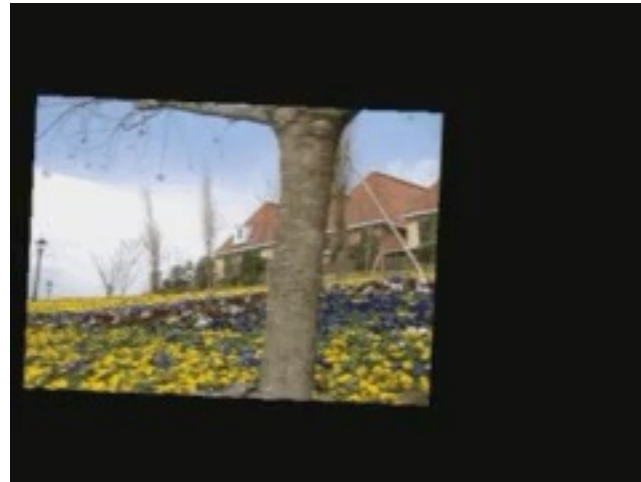
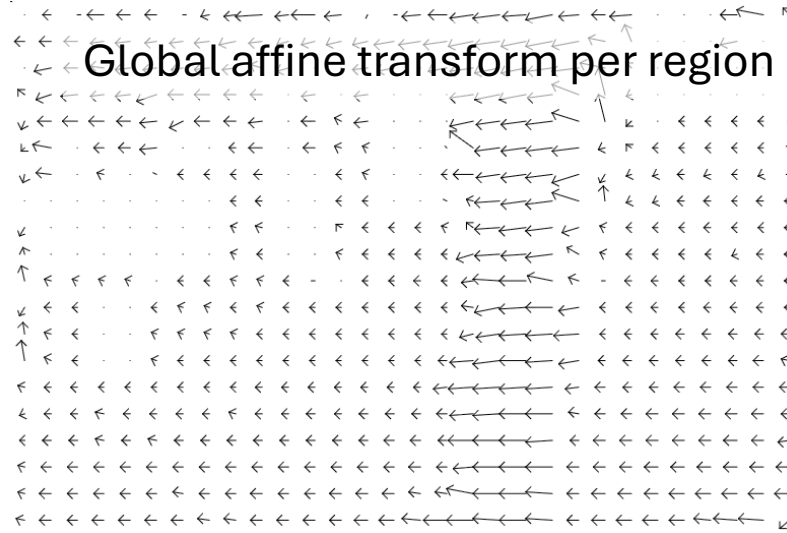
# Why not calc transform with key points?

Because we don't have them!

Don't know the model,  
Don't know the correspondences.

Finding the model that minimizes single pixel brightness constancy

# Motion segmentation





Massachusetts Institute of Technology



# Revealing Invisible Changes In The World

Created for the NSF International Science & Engineering  
Visualization Challenge 2012

# What breaks optical flow?

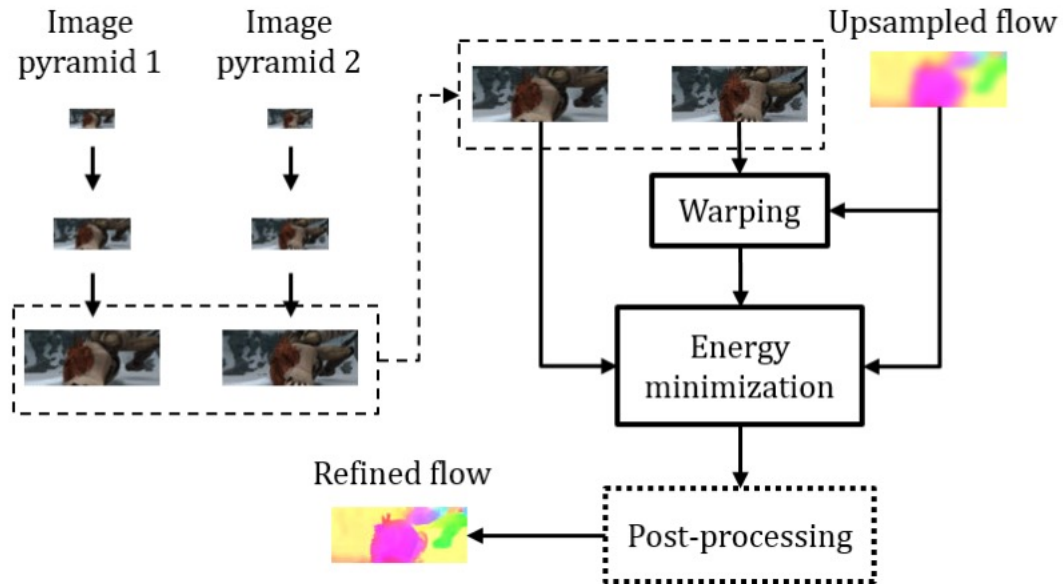
Or: Have we been naïve?

- Brightness constancy- does it hold?
- Occlusions
- 3D understanding
- Consistency across time

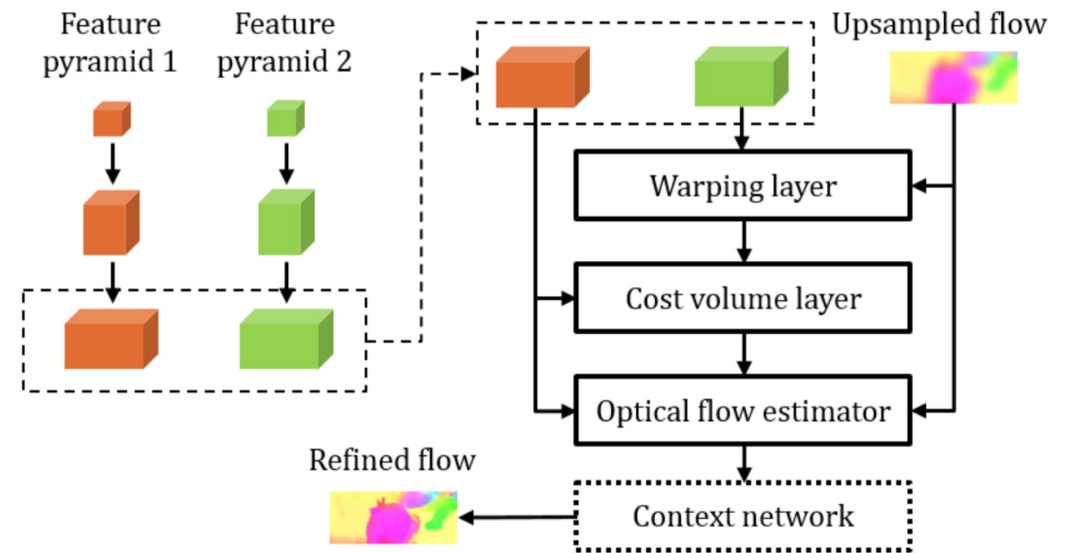


# Deep optical flow: track features

## Classic

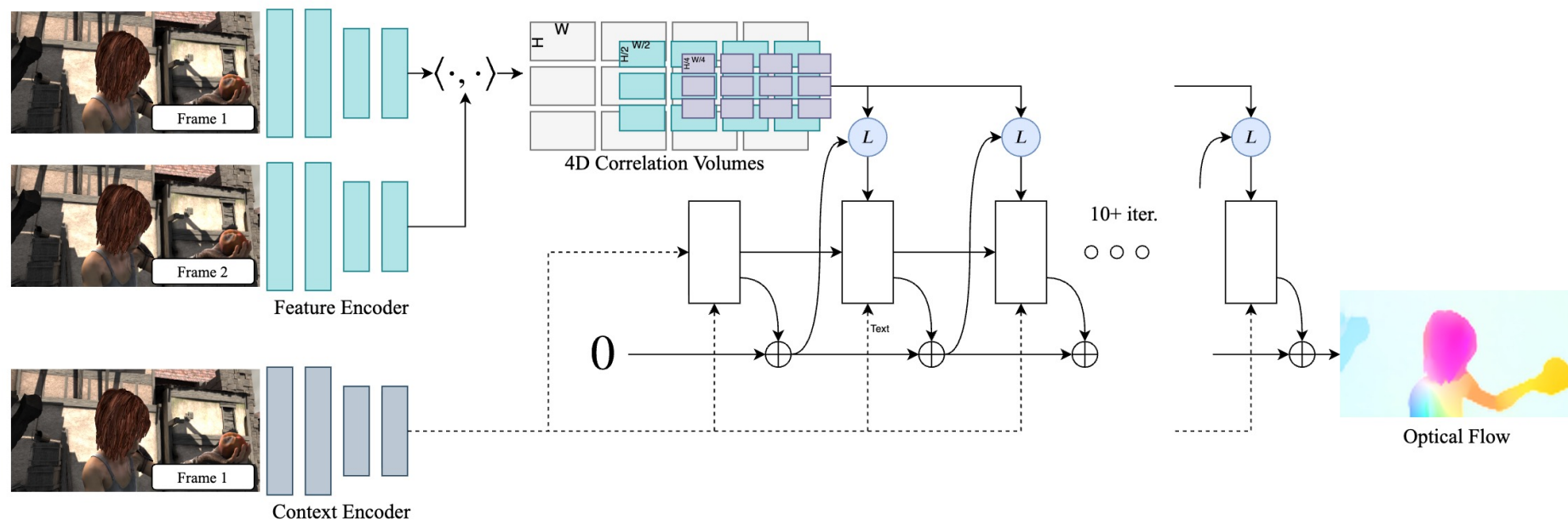


## Deep



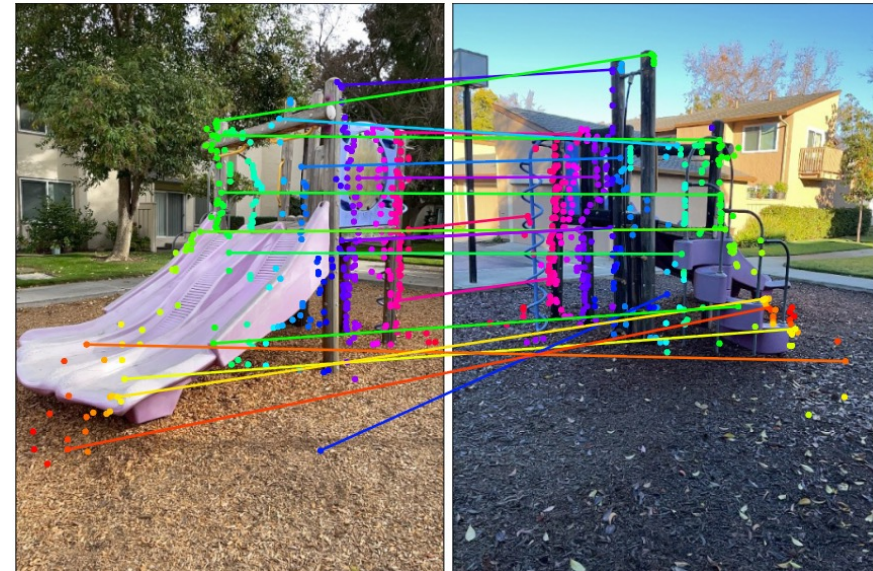
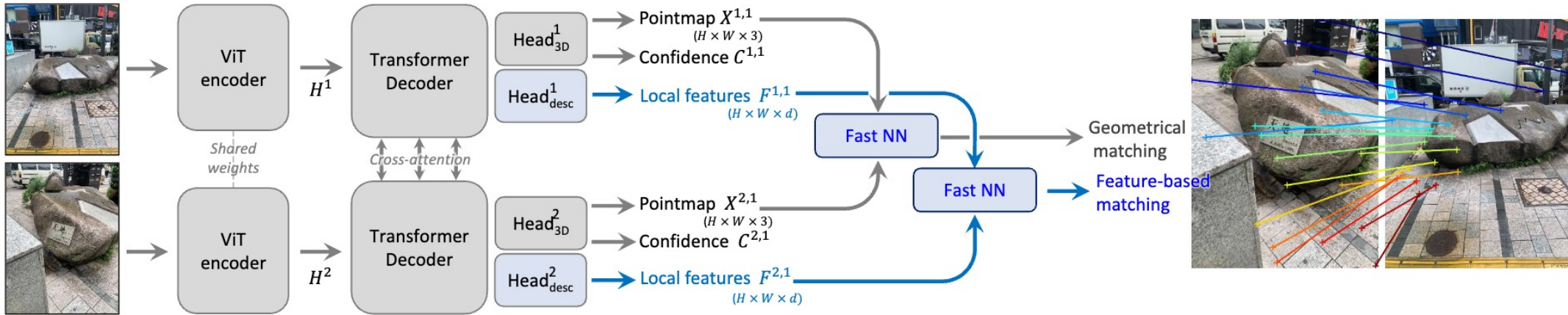
# RAFT

## Recurrent All-Pairs Field Transforms for Optical Flow

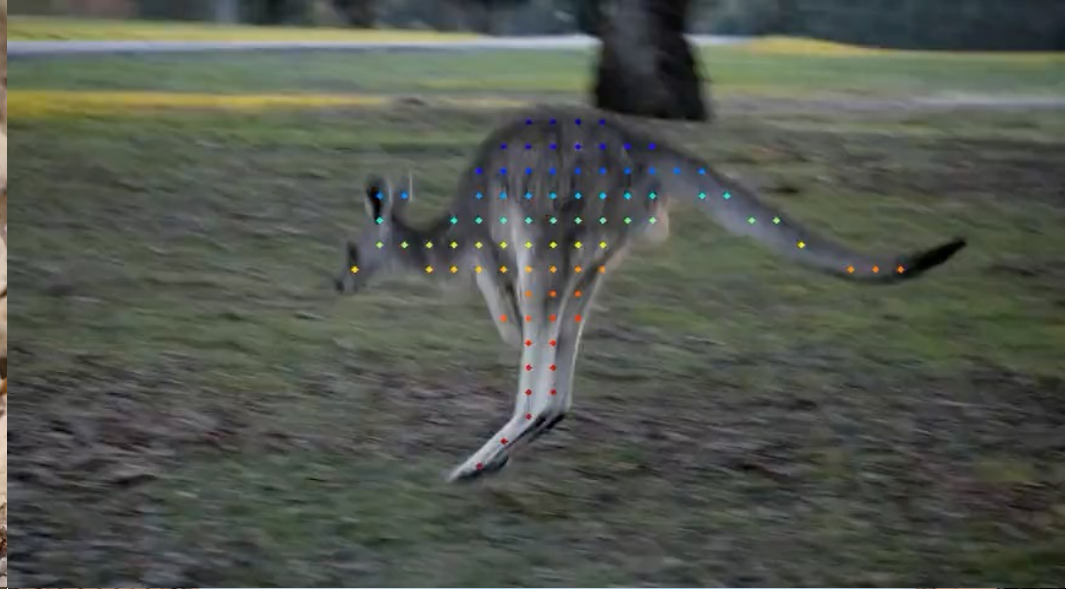


4D attention for feature matching

# Grounding Image Matching in 3D with MAST3R

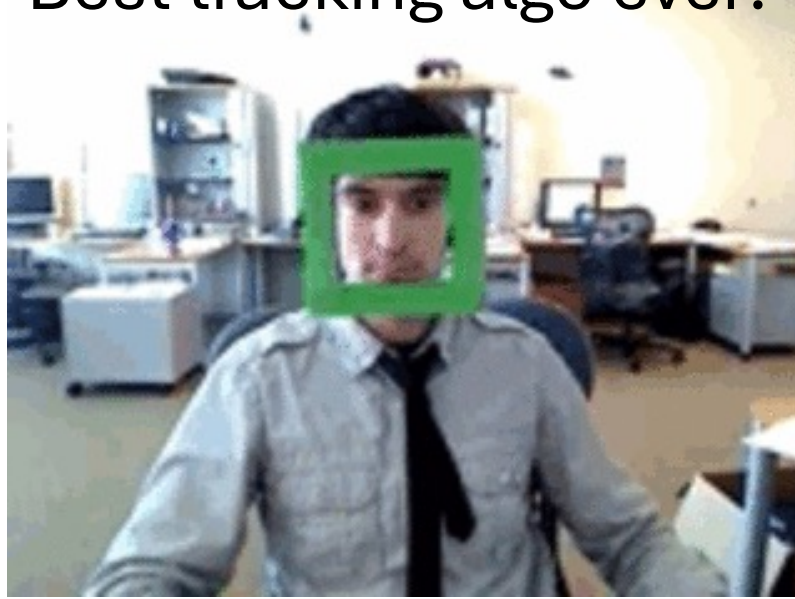


# Trajectory across time consistency



# Thank you!

Best tracking algo ever!



Next week:

## Generative models

